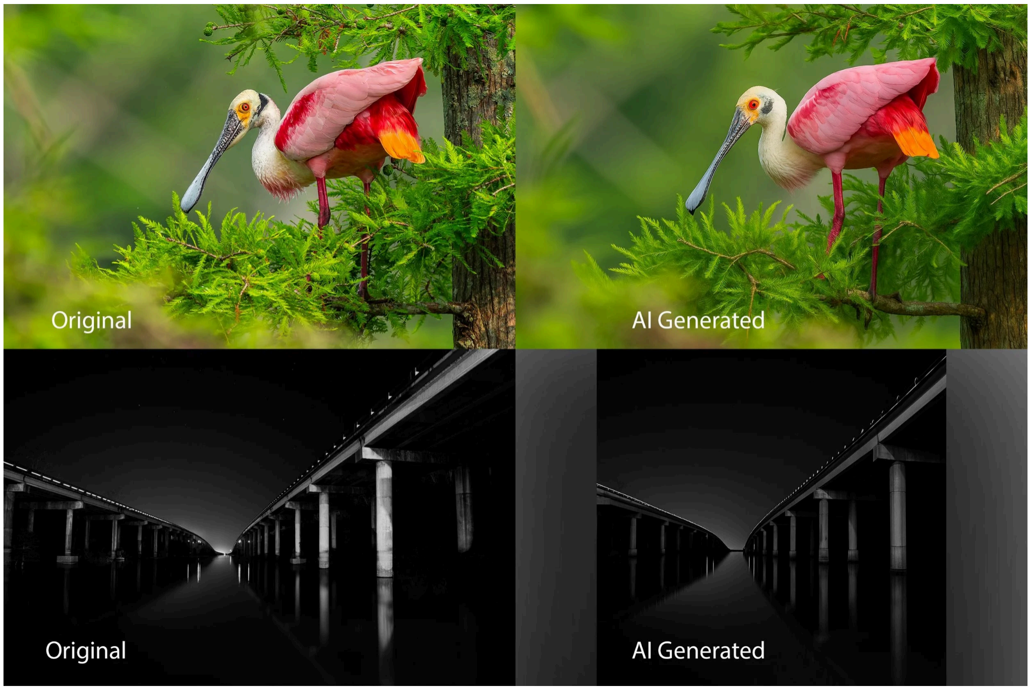
Can You Trust Your Eyes?

DS 4002 Case Study by Dev Patel



AI-generated imagery is everywhere. From social media feeds to news articles, synthetic images are being produced and shared at a scale that was unimaginable a few years ago. Tools capable of generating photorealistic images are now widely accessible, and the line between what is real and what is fake has never been blurrier. This presents a challenge for journalists, researchers, and other institutions that rely on media. This is where you come in.

You have been recruited as a junior data scientist at a digital media verification lab. You have been tasked with developing an automated pipeline to classify images as either real or AI-generated, supporting fact-checkers and content moderators who process thousands of images daily.

By using a publicly available dataset, you will design, train, and evaluate a Convolutional Neural Network (CNN). Your model will help identify a subtle visual pattern that distinguishes AI-generated images from authentic photographs. By doing so, you will not only help the lab and other institutions identify images better but also apply data science knowledge to real-world applications. For more details, visit this GitHub repository: https://github.com/devpp101/DS4002CS3.